

Cancer Support Netherlands

Patient Navigation Knowledge-Base Project

A research and data-product report on organising trusted Dutch cancer-care information for English-speaking international patients.

Executive summary

Cancer Support Netherlands (CSN) supports English-speaking international residents of the Netherlands affected by cancer. Although reliable Dutch healthcare information exists, it is fragmented across organisations and often unavailable in English. This project developed an information-retrieval pipeline that gathers public content, matches it to common patient-navigation questions, translates relevant passages, and exports a structured Excel dataset for review and future service development.

The project demonstrates a practical use of web scraping, semantic search, machine translation, and metadata enrichment. It does not provide medical advice and requires clinical and human review before any patient-facing use.

30 patient-navigation questions	30 structured output rows	0.661 mean similarity score	0.515-0.796 observed score range
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Why this matters

- Information access is a navigation challenge, not simply an information-availability problem.
- The dataset provides traceable source URLs, translations, answer summaries, and navigation metadata.
- The current results support a pilot knowledge base; they do not establish medical accuracy or complete answer relevance.

How the pipeline works

The pipeline compares each English patient question with paragraph-level content extracted from selected Dutch healthcare and cancer-information pages. Questions are translated to Dutch before matching to reduce language mismatch; the selected passage is then translated back into English for reporting.

1. Define questions 30 questions across referral, waiting times, treatment decisions, support, communication, orientation, and active navigation.	2. Collect public content Extract relevant paragraph-level text from selected Dutch healthcare webpages.
3. Retrieve by meaning Encode translated questions and paragraphs with all-MiniLM-L6-v2; select the highest cosine-similarity match.	4. Translate and enrich Translate matched text to English and add source, date, journey-stage, stakeholder, priority, and next-step fields.
5. Review and export Review scores and results; export the structured dataset to Excel for inspection and future development.	

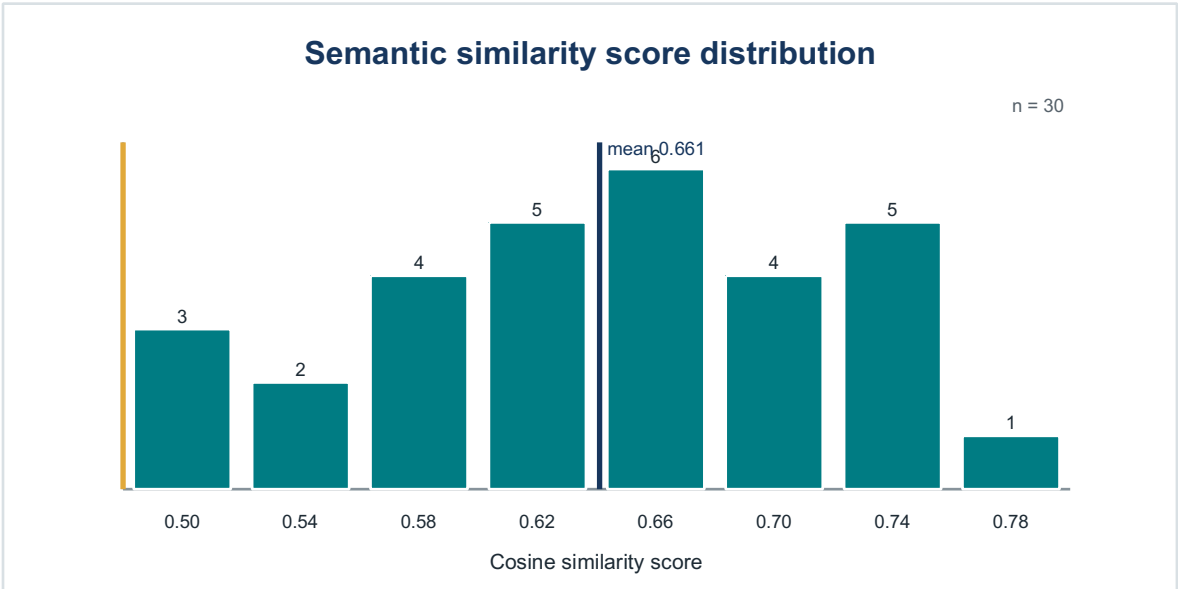
Dataset design

Question and topic	Original patient-navigation question and category.
Matched content	Dutch source paragraph, English translation, short answer summary, and similarity score.
Traceability	Source organisation, source-page title, URL, and scrape date.
Navigation metadata	Patient-journey stage, stakeholder category, priority, and suggested next step.

Source scope. The notebook refers to 29 selected public webpages from Dutch cancer and healthcare organisations, including Kanker.nl, Thuisarts.nl, Zorgwijzer.nl, AVL, and Teladoc Health. Web content can change; retained source URLs and collection dates are therefore essential.

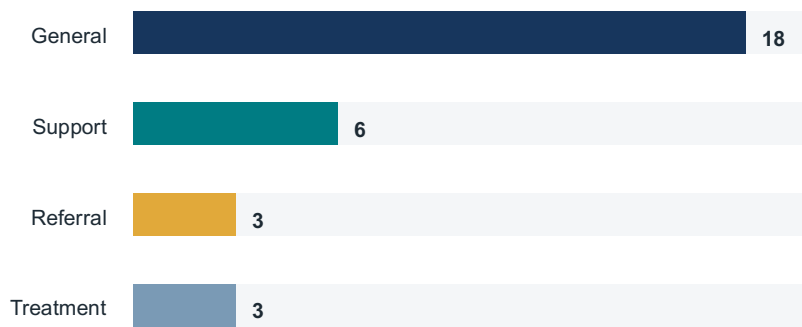
Results and evidence

All 30 questions produced an output row. Similarity scores were concentrated between 0.515 and 0.796, with a mean of 0.661. The score indicates semantic proximity between a question and a retrieved paragraph; it is not a measure of clinical correctness or a replacement for manual relevance review.



Interpretation. The current priority rule labels scores of 0.50 or above as "High". Since every observed score exceeds 0.50, all 30 records received a High label. The threshold is therefore not sufficiently discriminating for quality assurance and should be recalibrated after human evaluation.

Patient-navigation questions by journey stage



Number of questions

Interpretation. Most questions are classified as General, while support, referral, and treatment are represented in smaller numbers. The imbalance is partly driven by the project's rule-based classification and should not be interpreted as a population-level measure of patient need.

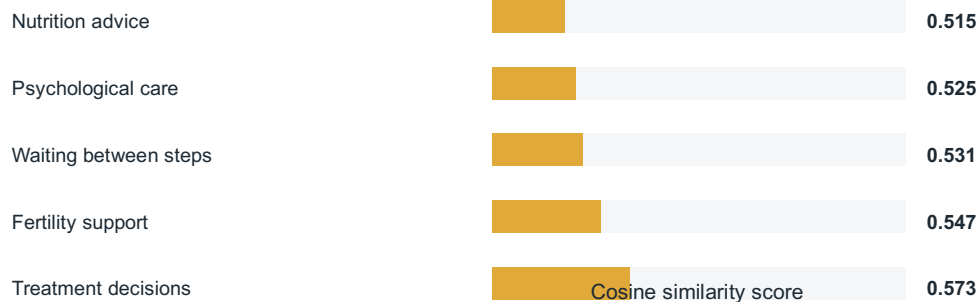
Quality review, recommendations, and safeguards

Highest and lowest retrieval scores

Highest scoring matches



Lowest scoring matches



The range chart identifies examples for manual inspection. High scores can still produce only partially relevant answers, while low scores may flag gaps in source coverage, question wording, translation, or retrieval performance. A future version should add a human relevance label for every match and report precision at a chosen threshold.

Recommendations

- **Introduce human review.** Create a relevance rubric and ask qualified reviewers to label whether the retrieved passage fully, partially, or does not answer each question.
- **Recalibrate priority.** Replace the current single 0.50 threshold with evidence-based bands derived from manual relevance labels.
- **Expand coverage carefully.** Add cancer-specific pages and trusted sources for insurance, medical terminology, and care pathways, while recording permissions and update dates.
- **Prepare for safe deployment.** Any patient-facing navigator should show source links, state its limitations, include escalation to a healthcare professional, and be governed by clinical review.

Limitations

- The project uses a small, curated question set and selected public webpages; it is not a comprehensive representation of all patient needs or Dutch cancer-care content.
- Web scraping must comply with each website's terms of use, robots.txt guidance, and rate limits.
- Machine translation, automated summaries, and semantic matching can introduce errors. No output should be presented as clinical advice.

Prepared by Solomon Alfred | September 2026